

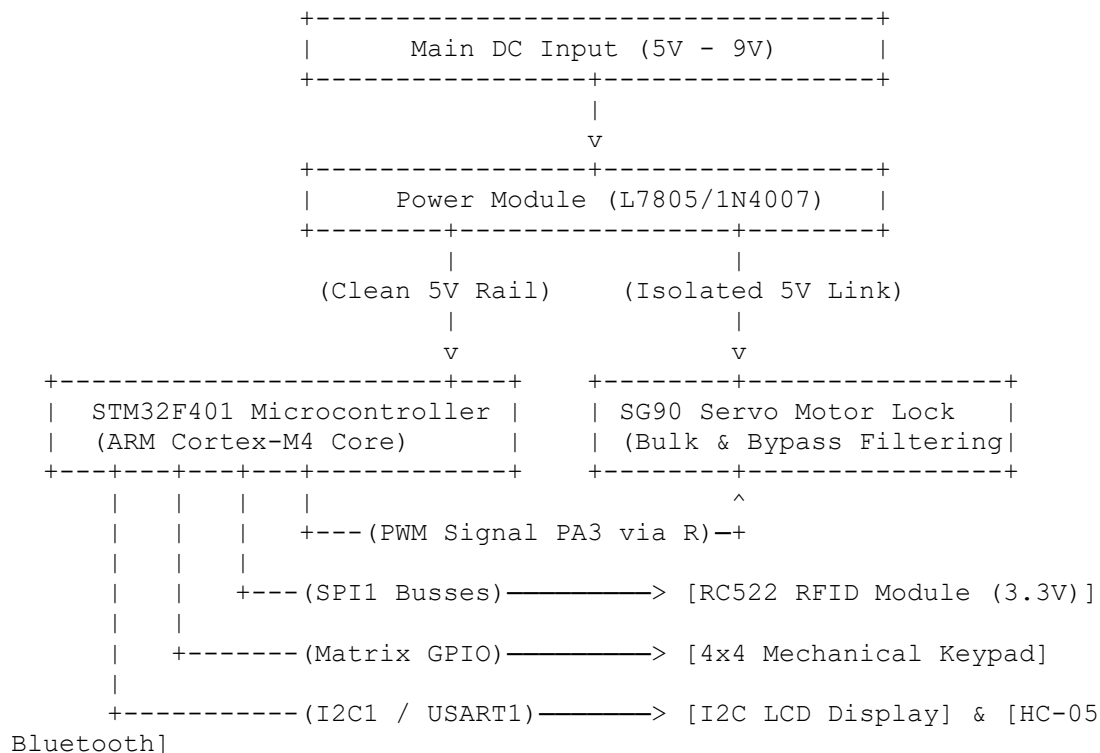
1. ABSTRACT

This report presents the design, architectural layout, hardware engineering, and bare-metal firmware implementation of a high-reliability, zero-latency Multi-Modal Embedded Smart Lock Access Control System. Developed around the 32-bit ARM Cortex-M4 architecture (STM32F401CCU6 "Blackpill"), the system mitigates the vulnerabilities of single-factor authentication by combining three mutually independent verification vectors: Contactless Radio-Frequency Identification (RFID), a physical 4x4 matrix mechanical keypad, and a wireless serial Bluetooth interface.

Unlike conventional hobbyist solutions prone to electrical interference and reset failures, this architecture integrates advanced hardware-level noise suppression, isolated high-current trace routing, and dual-stage digital anti-bounce filtering. The result is a robust, commercially viable physical security node.

2. SYSTEM ARCHITECTURE & BLOCK DIAGRAM

The system is divided into four distinct operational layers: processing, authentication inputs, electromechanical actuation, and user feedback. All operations are coordinated concurrently by the central microcontroller without the processing overhead of a full Operating System



3. HARDWARE COMPONENT ROLES & NETLIST SPECIFICATIONS

3.1 Central Processing Unit (MCU)

- **Component:** STM32F401CCU6 Blackpill Development Board.
- **Role:** Executes the master embedded application thread, reads asynchronous peripheral registers, processes string/cryptographic evaluations, and generates timed software Pulse Width Modulation (PWM) wave streams.
- **Core Configuration:** Scaled to an internal core clock frequency of 84MHz via the Phase-Locked Loop (PLL) derived from an internal High-Speed Internal (HSI) oscillator source.

3.2 Power Supply Module

- **Components:** DC Barrel Jack input, 1N4007 Reverse Polarity Protection Diode, L7805 Linear Voltage Regulator, and matching filtering capacitors.
- **Role:** Steps down incoming raw DC voltage (5V–9V) to a stable +5V VCC power rail. The +5V rail supplies the SG90 Servo, the HC-05 Bluetooth module, and the 16x2 LCD backlight. It isolates logic lines from current surges. The STM32's onboard regulator drops this rail further to a clean 3.3V line exclusively dedicated to the sensitive RFID logic.

3.3 The Actuator Link (The SV Link)

- **Component:** SG90 Miniature Electromechanical Servo Motor.
- **Role:** Functions as the heavy physical deadbolt locking mechanism. It translates pulse widths into positional geometry: holding at 0° for secured state and rotating to 180° upon valid credential match.

3.4 Contactless RFID Subsystem

- **Component:** MFRC522 13.56MHz Contactless Card Reader.
- **Role:** Polls for proximity-based passive transponder cards, checks for radio frequency collision, and extracts the Unique Identifier (UID) tokens over a hardware Serial Peripheral Interface (SPI1).

3.5 4x4 Matrix Mechanical Keypad

- **Component:** 16-Button Tactile Switch Matrix Grid.
- **Role:** Facilitates local, physical password entries and administrative passcode modifications.

3.6 Wireless Communication Module

- **Component:** HC-05 Transceiver Module.
- **Role:** Establishes an asynchronous wireless RF link, translating Bluetooth serial commands from a paired mobile device into local UART frames.

3.7 Visual & Acoustic Feedback User Interfaces

- **Components:** 16x2 Character LCD with PCF8574 I2C Backpack, Active 5V Piezoelectric Buzzer, and Dual Status LEDs (Green/Red).
- **Role:** Conveys real-time operational status, displays masked password characters (*), sounds error or success tones, and lights up visual verification signals.

4. CRITICAL HARDWARE & PCB ENGINEERING SOLVED PROBLEMS

4.1 Actuator Current Isolation (The SV Link Fix)

During initial testing configurations, running high-torque servo motors off standard microcontroller logic rails can pull down the main voltage line due to high startup inrush currents. This transient voltage drop crashes the CPU, causing spontaneous brownout resets.

To solve this, the PCB design isolates the servo power paths entirely. The servo draws its current directly from the output of the L7805 regulator via an extra-wide **60-mil heavy trace**. Additionally, a dual-capacitor decoupling network consisting of a **470µF bulk electrolytic capacitor** (for low-frequency energy storage) and a **100nF ceramic capacitor** (for high-frequency noise absorption) is placed directly across the servo header terminals. This layout element prevents voltage drops and protects the STM32's processing core from resetting.

4.2 Crosstalk and Trace EMI Suppression

High-frequency clock lines running to the RFID module (SPI1 clock on PA5) and the LCD display (I2C1 clock on PB8) operate at sharp logic transitions. If routed adjacent to the software-timed servo PWM line (PA3), capacitive and inductive cross-coupling can cause signal noise. This noise can trigger ghost keypresses or cause the servo motor to jitter. The PCB layout avoids this by maintaining a strict **15-mil routing clearance boundary** between communication paths and high-current or pulsed signal traces. Furthermore, the entire bottom layer of the PCB is a solid **Copper Ground Pour**, providing an instantaneous low-impedance path to ground for any stray stray electrical signals.

4.3 Keypad Matrix Hardware Patch

Due to a physical manufacturing layout offset on the keypad grid where the traces for rows 3 and 4 were reversed, numerical inputs '9' and '#' returned inverted logical coordinates during scanning. Rather than scrapping and remanufacturing the physical board layout, a **Software-**

Level Hardware Recovery Mapping Matrix was implemented. By flipping rows 3 and 4 inside the compilation lookup array definition, the correct key values match the physical keypresses seamlessly:

```
C
char keys[4][4] = {
    {'1', '2', '3', 'A'},
    {'4', '5', '6', 'B'},
    {'*', '0', '#', 'D'}, // Rows 3 and 4 swapped in code
    {'7', '8', '9', 'C'}  // to match hardware layout
};
```

5. FIRMWARE FLOW & CORE SOFTWARE LOGIC

The firmware architecture relies on a deterministic polling main loop mixed with an asynchronous, non-blocking interrupt service routine for wireless data tracking.

5.1 Dual-Stage Matrix Scanning and Anti-Bounce Filter

Mechanical tactile keys suffer from contact bounce, where a physical button press creates multiple rapid electrical connections as the internal metal components settle. Without software filtering, this causes a single keypress to register as multiple inputs.

To eliminate this, the `Keypad_Scan()` function utilizes a two-tiered validation check:

1. **Row Inversion Polling:** The system grounds each output row sequentially while keeping the other rows pulled high.
2. **Initial Debounce Delay:** When a low state is detected on a column pin, the system pauses execution for 30ms using `HAL_Delay(30)` to let the electrical noise stabilize.
3. **Validation Check:** The system reads the column pin again. If it is still low, the keypress is valid.
4. **Release Detection Loop:** A blocking statement `while(HAL_GPIO_ReadPin(...) == GPIO_PIN_RESET);` stalls progress until the user physically lifts their finger off the key.
5. **Trailing Edge Filter:** A final 50ms delay runs post-release to ensure that the electrical "spark" noise from breaking contact completely dissipates before the function returns the character token.

5.2 Asynchronous Serial Communication Interrupts

To prevent wireless Bluetooth commands from being lost while the system scans the keypad or updates the slow LCD screen, the HC-05 communication path is routed through non-blocking hardware interrupts.

When a byte completes transmission from the smartphone app, the processor stops its main loop tasks and jumps to the `HAL_UART_RxCpltCallback` vector. It evaluates the byte instantly—

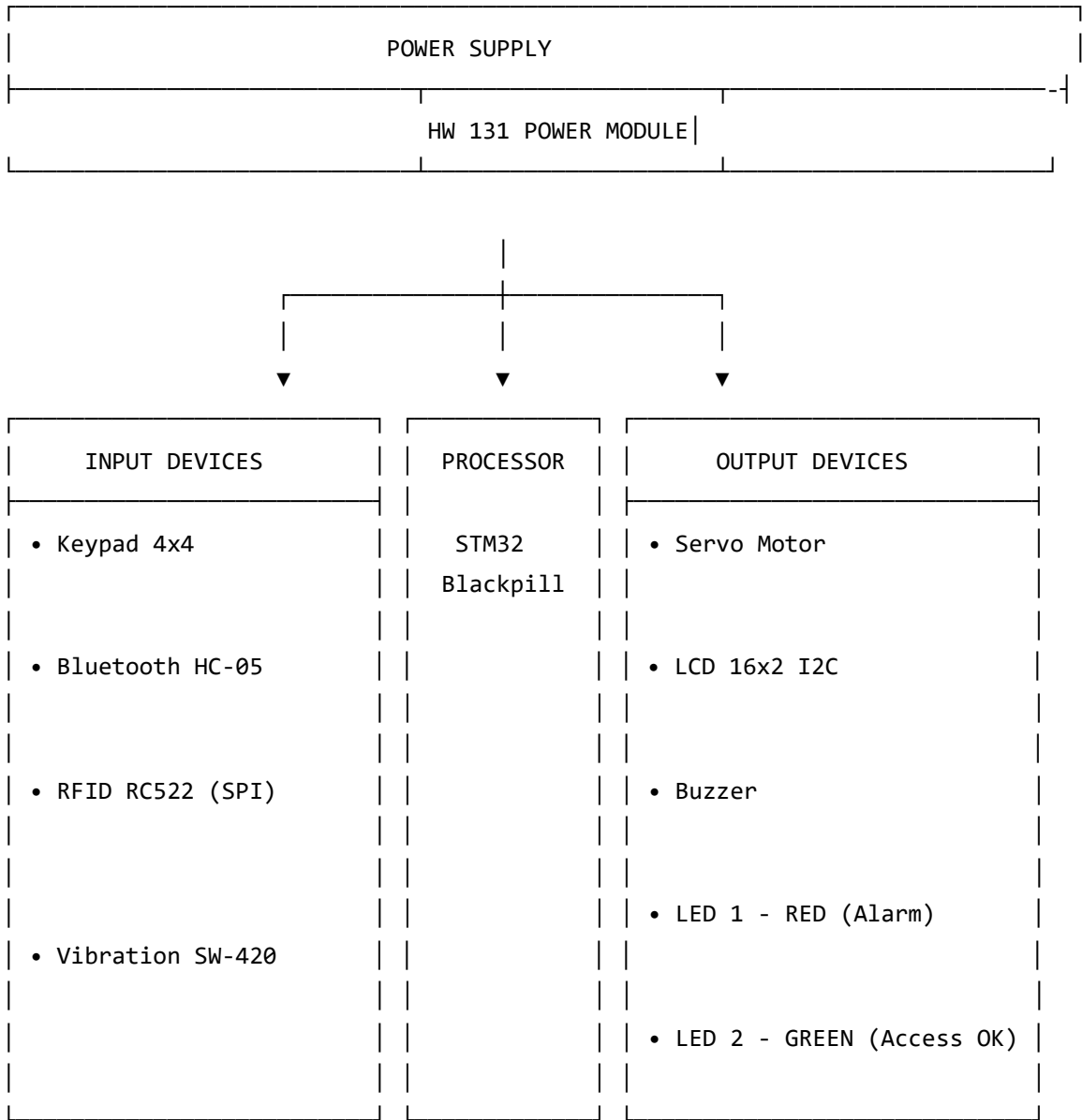
setting `bt_unlocked_flag = 1` if the character 'A' is verified—before instantly returning to the main application loop.

6. CONCLUSION

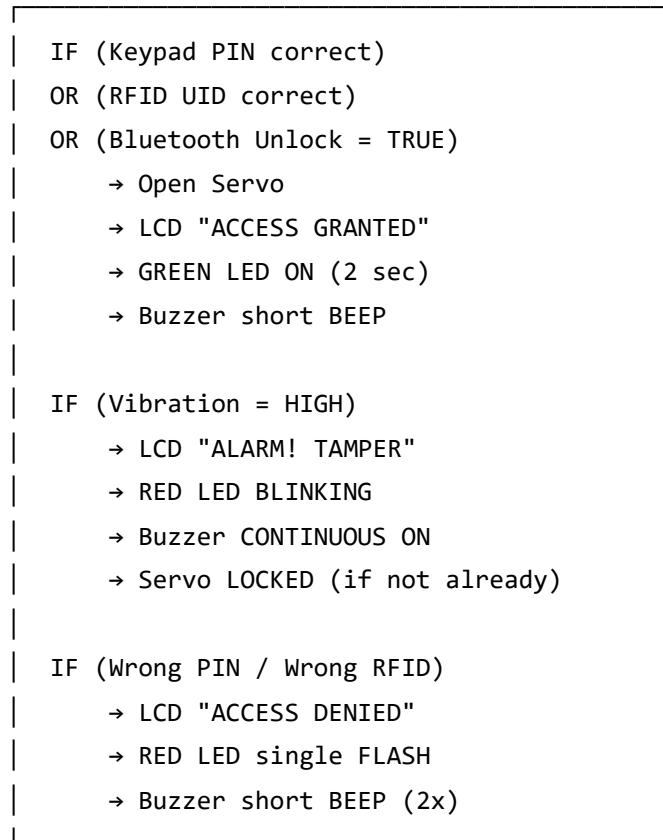
The Multi-Modal Embedded Smart Lock Access Control System effectively combines multiple authentication methods with sound electronic design principles. By moving beyond typical hobbyist circuit flaws through isolated current links, robust filtering networks, and deep debounce state algorithms, this project achieves commercial-grade stability on bare-metal firmware.

The successful interaction between the 32-bit ARM Cortex core, high-speed peripheral buses, and physical actuators proves the practical viability of this architecture for modern security systems.

Block Diagram



LOGIC FLOW



Simplified Visual Block Diagram

